

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2018-2019)

PRE-MEDICAL : NURTURE COURSE PHASE-III

TEST # 03

DATE : 25-08-2018

Test Type : Minor

Test Pattern : AIIMS

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **180**.
5. **Each correct answer carry 1 mark. While one third mark will be deducted for each incorrect answer.**
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

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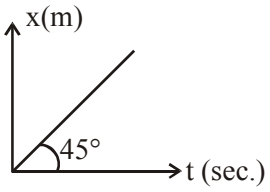
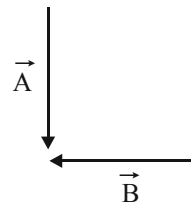
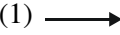
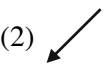
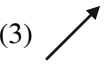
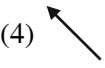
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Your Target is to secure Good Rank in AIIMS 2020

MINOR TEST # 03		Topics	DATE : 25 - 08 - 2018
PRE-MEDICAL : NURTURE COURSE PHASE-III			
PHYSICS	:	Basic Maths, Vectors, Kinematics, Projectile Motion	
CHEMISTRY	:	Mole Concept, Atomic Structure, Periodic Table	
BIOLOGY	:	Cell Biology, Animal Tissue, Biomolecule	

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

- When a clock shows 5 o'clock, how much angle do its minute and hour needles make :-
 (1) 90° (2) 120°
 (3) 135° (4) 150°
- Given $\operatorname{cosec} \theta = 4$. Find the value of $\tan \theta$. If θ lies in the first quadrant :-
 (1) $\frac{1}{4}$ (2) $\frac{1}{\sqrt{17}}$
 (3) $\frac{1}{\sqrt{15}}$ (4) $\sqrt{15}$
- Value of $\tan(-60^\circ)$ is :-
 (1) $\frac{1}{\sqrt{3}}$ (2) $-\frac{1}{\sqrt{3}}$
 (3) $\sqrt{3}$ (4) $-\sqrt{3}$
- The greatest value of function $6\sin \theta - 8\cos \theta$ is :-
 (1) 6 (2) 8 (3) 10 (4) 14
- Approximate value of $\tan 3^\circ$ is :-
 (1) 3 (2) $\frac{\pi}{180}$ (3) $\frac{\pi}{60}$ (4) $\frac{\pi}{30}$
- Position time graph of particle is shown in figure. Then velocity of particle is :-

 (1) 0 (2) $\frac{1}{\sqrt{2}}$ m/sec
 (3) 1 m/sec (4) 2 m/sec
- If position of particle is given by $x = 2t^2 - 8t + 16$ where x in meter and t in sec. Then position of particle when it comes to rest :-
 (1) 2m (2) 4m
 (3) 8m (4) 16m
- If $x = t^3 + 2t^2 - 4t + 1$ where t is in second and x is in meter then initial acceleration is :-
 (1) 2 m sec^{-2} (2) 4 m sec^{-2}
 (3) 6 m sec^{-2} (4) 8 m sec^{-2}
- Value of $\sin x + \cos x$ is maximum at :-
 (1) $x = 0$ (2) $x = \frac{\pi}{4}$
 (3) $x = \frac{\pi}{3}$ (4) $x = \frac{\pi}{2}$
- If $\vec{F}_1 = 500\text{N } \hat{k}$ and $\vec{F}_2 = 500\text{N } \hat{j}$ then which statement is correct.
 (1) $\vec{F}_1 \neq \vec{F}_2$ (2) $\vec{F}_1 \perp \vec{F}_2$
 (3) $|\hat{F}_1| = |\hat{F}_2|$ (4) all of the above
- Which of following statements is/are correct :-
 (1) angle between parallel vectors is zero
 (2) angle between antiparallel vectors is 180°
 (3) parallel vectors have same unit vector and antiparallel vectors have opposite unit vector
 (4) All
- If $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$, then angle between $\vec{A}$ and $\vec{B}$ will be :-
 (1) 0° (2) 90°
 (3) 180° (4) 45°
- If magnitude of two vectors is 4 and 8, which of following cannot be its resultant :-
 (1) 5 (2) 10
 (3) 15 (4) All
- If two vectors are given as

 then $\vec{A} + \vec{B}$ will be :-
 (1)  (2) 
 (3)  (4) 

15. $\vec{A} = 2\hat{i} + 3\hat{j} + 4\hat{k}$

$\vec{B} = \hat{i} - \hat{j} + \hat{k}$

Find unit vector along $\vec{A} + \vec{B}$.

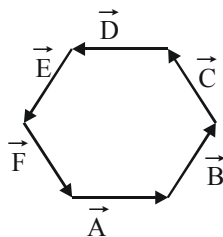
(1) $\frac{-3\hat{i} + 2\hat{j} + 5\hat{k}}{\sqrt{38}}$

(2) $\frac{-\hat{i} - \hat{j} - \hat{k}}{\sqrt{3}}$

(3) $\frac{3\hat{i} + 2\hat{j} + 5\hat{k}}{\sqrt{38}}$

(4) None

16.



$\vec{A} + \vec{B} + \vec{C} + \vec{D} + \vec{E} + \vec{F} =$

(1) 0

(2) $\vec{0}$

(3) $6\vec{A}$

(4) All

17. The magnitude of resultant of the two vectors of magnitude 4 unit and 3 unit is 1 unit. The angle between them in radian is :-

(1) 0

(2) $\frac{\pi}{4}$

(3) $\frac{\pi}{2}$

(4) π

18. How many minimum number of coplanar vectors having different magnitudes can be added to give zero resultant :-

(1) 2

(2) 3

(3) 4

(4) 5

19. What happens when we multiply a vector by 2.

(1) direction reverse and magnitude double

(2) direction remain unchanged and magnitude double

(3) direction and magnitude remain unchanged.

(4) None

20. If $\vec{A} = 5\hat{i} + 10\hat{j} - \hat{k}$ and $\vec{A} + \vec{B}$ is a unit vector along x axis then $\vec{B}$ is equal to :-

(1) $\hat{i}$

(2) $-4\hat{i} - 10\hat{j} + \hat{k}$

(3) $4\hat{i} + 10\hat{j} - \hat{k}$

(4) None of these

21. The horizontal range and maximum height attained by a projectile are R and H respectively. If a constant horizontal acceleration $a = g/4$ is imparted to the projectile due to wind, then its horizontal range and maximum height will be :-

(1) $(R + H), \frac{H}{2}$

(2) $\left[R + \frac{H}{2}\right], 2H$

(3) $(R + 2H), H$

(4) $(R + H), H$

22. A pebble is released from rest at a certain height and falls freely, reaching an impact speed of 4 m/s at the floor. Next, the pebble is thrown down with an initial speed of 3 m/s from the same height, which is the speed at the floor ?

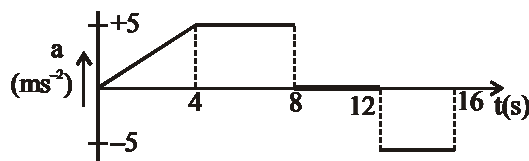
(1) 4 m/s

(2) 5 m/s

(3) 6 m/s

(4) 7 m/s

23. The acceleration of a particle traveling along a straight line is shown in the fig. The maximum speed of the particle is particle starts from rest :-



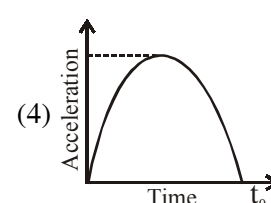
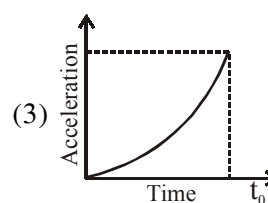
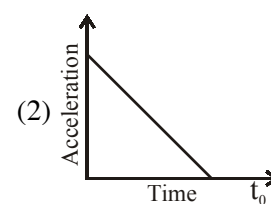
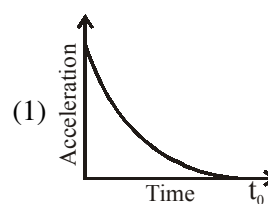
(1) 20 ms^{-1}

(2) 30 ms^{-1}

(3) 40 ms^{-1}

(4) 60 ms^{-1}

24. Acceleration versus time graphs for four objects are shown below. All axes have the same scale. Which object had the greatest change in velocity during the interval ?



25. The trajectory of a projectile in a vertical plane is $y = ax - bx^2$, where a and b are constant and x and y are respectively horizontal and vertical distance of the projectile from the point of projection. The maximum height attained by the particle and the angle of projection from the horizontal are :-

- (1) $\frac{b^2}{2a}, \tan^{-1}(b)$ (2) $\frac{a^2}{b}, \tan^{-1}(2b)$
(3) $\frac{a^2}{4b}, \tan^{-1}(a)$ (4) $\frac{2a^2}{b}, \tan^{-1}(a)$

26. The speed of a projectile at its maximum height is $\sqrt{3}/2$ times its initial speed. If the range of the projectile is n times the maximum height attained by it, n is equal to :-

- (1) $\frac{4}{3}$ (2) $2\sqrt{3}$ (3) $4\sqrt{3}$ (4) $\frac{3}{4}$

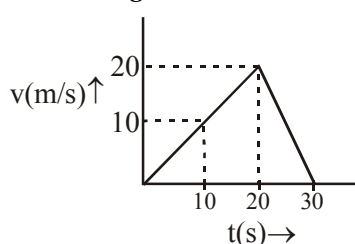
27. A body of mass m thrown horizontally with velocity v , from the top of tower of height h touches the level ground at a distance of 250 m from the foot of the tower. A body of mass 2 m thrown horizontally with velocity $v/2$, from the top of tower of height $4h$ will touch the level ground at a distance x from the foot of tower. The value of x is :-

- (1) 250 m (2) 500 m
(3) 125 m (4) $250\sqrt{2}$ m

28. From the top of a tower of height 40 m, a ball is projected upwards with a speed of 20 ms^{-1} at an angle of elevation of 30° . Then the ratio of the total time taken by the ball to hit the ground to the time to ball come at same level as top of tower :-

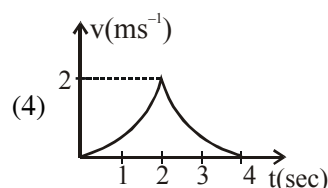
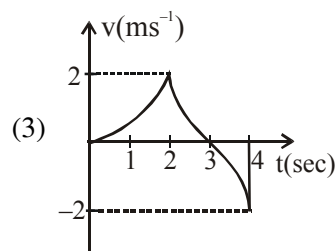
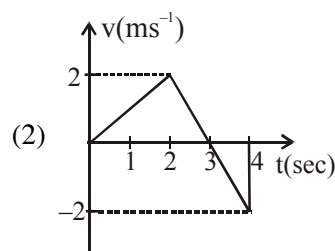
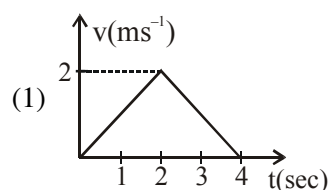
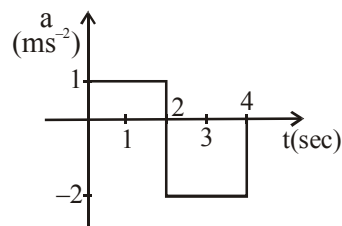
- (1) 2 : 1 (2) 3 : 1 (3) 3 : 2 (4) 4 : 1

29. V-t graph of an object of mass 1 kg is as shown. Select the **wrong** statement



- (1) Distance and displacement are equal.
(2) the average acceleration of the object is zero
(3) the average velocity of object is zero
(4) the acceleration acting on particle at $t = 10$ sec is 1 m/s^2

30. A particle moves in one dimension. Its velocity is given by v and its acceleration by a . The figure shows its measured acceleration versus time graph. Which of the graph (v vs t) below are consistent with measured acceleration?



31. Each of four particles move along x -axis. Their coordinates (in meters) as functions of time (in seconds) are given by
particle 1 : $x(t) = 3.5 - 2.7t^3$
particle 2 : $x(t) = 3.5 + 2.7t^3$
particle 3 : $x(t) = 3.5 + 2.7t^2$
particle 4 : $x(t) = 3.5 - 3.4t - 2.7t^2$
Which of these particles have constant acceleration ?

- (1) All four
(2) Only 1 and 2
(3) Only 2 and 3
(4) Only 3 and 4

32. Height of a magical tree is increasing with time t as $h = t^2 + 2t + 5$ m then rate of change of height at time $t = 2$ sec.

- (1) 2 m/sec
(2) 4 m/sec
(3) 6 m/sec
(4) 13 m/sec

33. If $y = \sin x + \sqrt{3} \cos x$ then for what value of x , y is maximum

- (1) 0° (2) 30°
(3) 45° (4) 60°

34. Co-ordinates of two particles is given by

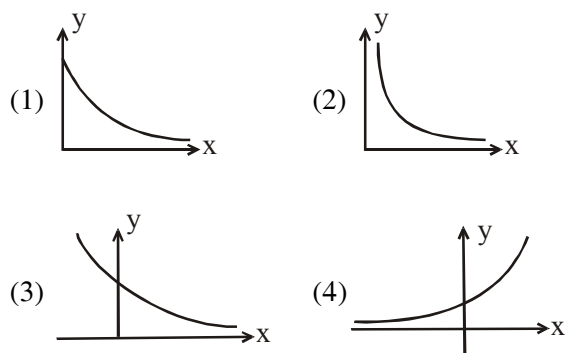
$$x_A = 3t \quad x_B = 15t$$

$$y_A = 2t^2 \quad y_B = 7t^2$$

Then slope of line AB at time $t = 1$ sec will be :

- (1) $\frac{3}{4}$ (2) $\frac{4}{3}$
(3) $\frac{5}{12}$ (4) $\frac{5}{13}$

35. Graph of $y = e^x$ will be :-



36. If position of a particle is $x = t^2 - 10t + 60$ m then find the position of particle at the time when it comes to rest.

- (1) 5 m
(2) 25 m
(3) 35 m
(4) 40 m

37. Two balls are projected at an angle θ and $(90^\circ - \theta)$ to the horizontal with the same speed. The ratio of their maximum vertical heights is:-

- (1) 1 : 1
(2) $\tan \theta : 1$
(3) $1 : \tan \theta$
(4) $\tan^2 \theta : 1$

38. For angles of projection of a projectile at angles $(45^\circ - \theta)$ and $(45^\circ + \theta)$, the horizontal ranges described by the projectile are in the ratio of:-

- (1) 1 : 1
(2) 2 : 3
(3) 1 : 2
(4) 2 : 1

39. A particle moving with a speed v changes direction by an angle θ , without change in speed :-

- (1) The change in the magnitude of its velocity

$$\text{is } 2v \sin \frac{\theta}{2}$$

- (2) The change in the magnitude of its velocity is zero

- (3) The magnitude of change in velocity is $v(1 - \cos \theta)$

- (4) The magnitude of change in velocity $\frac{v}{2}$

40. Two adjacent sides of a parallelogram represented by the two vectors $\hat{i} + 2\hat{j} + 3\hat{k}$ and $3\hat{i} - 2\hat{j} + \hat{k}$. What is the area of parallelogram?

- (1) 8 (2) $8\sqrt{3}$
(3) $3\sqrt{8}$ (4) 192

41. Elements with atomic number 47 belongs to -
 (1) 4th period & XII group
 (2) 4th period & XI group
 (3) 5th period & XII group
 (4) 5th period & XI group
42. Value of IE of Na ($Z = 11$) would be -
 (1) Greater than the IE of Ne
 (2) Below the IE of Ne but above the IE of oxygen
 (3) Lesser than the IE of Li
 (4) Between the IE of N and O.
43. Which of the following pairs has both members from the same group of periodic table -
 (1) Mg, Ba (2) Mg, Na
 (3) Mg, Cu (4) Mg, Cl
44. Select the elements of III A group of the periodic table. (Atomic no. are given) -
 (1) 3, 11, 19, 37
 (2) 5, 13, 21, 39
 (3) 7, 15, 31, 49
 (4) 5, 13, 31, 49
45. The law of triads was given by -
 (1) Mendeleeff
 (2) Newlands
 (3) Lothar Meyer
 (4) Dobereiner
46. Alkaline earth metals belong to which group -
 (1) IA (2) IIA
 (3) IIIA (4) IVA
47. Atomic number 25 element is belong to which block -
 (1) s-block (2) p-block
 (3) d-block (4) All of them
48. In which of the following element has highest value of I. P.
 (1) N (2) O
 (3) Ne (4) None of these
49. In C, N, O, F which of the following order is correct for I. P. : -
 (1) $F > O > C > N$
 (2) $O > F > N > C$
 (3) $F > N > O > C$
 (4) $N > F > O > C$
50. In which of the following have higher difference in the value of IInd and IIIrd I.P. -
 (1) $1s^2 2s^2 2p^6 3s^2 3p^1$
 (2) $1s^2 2s^2 2p^6 3s^2$
 (3) $1s^2 2s^2 2p^6 3s^1$
 (4) $1s^2 2s^2 2p^6 3s^2 3p^2$
51. Which of the following has highest metallic character -
- | Element | I. P. |
|---------|---------|
| (1) A | 16.2 eV |
| (2) B | 1.5 eV |
| (3) C | 12.2 eV |
| (4) D | 10.5 eV |
52. The energy required to remove an electron of an isolated gaseous atom from its ground state is called
 (1) Potential energy
 (2) Ionization energy
 (3) Electrode potential
 (4) Activation energy
53. In the long form of periodic table, the element having lowest ionisation potentials are present in :
 (1) I group (2) IV group
 (3) VII group (4) Zero group
54. Which of the following elements will have the lowest first ionisation energy ?
 (1) Mg (2) Rb
 (3) Li (4) Ca

- 55.** The ionization energy of nitrogen is more than that of oxygen because :
- Of the extra stability of half-filled p-orbitals in nitrogen
 - Of the smaller size of nitrogen
 - The former contains less number of electrons
 - The former is less electronegative
- 56.** Which is the correct order of electron affinity-
- O > S > Se > Te
 - S > O > Se > Te
 - S > Se > Te > O
 - O > Se > Te > S
- 57.** Which of the following element has the highest value of electron affinity -
- Carbon
 - Oxygen
 - Fluorine
 - Neon
- 58.** Which of the following element has the lowest value of electron affinity -
- Fluorine
 - Chlorine
 - Bromine
 - Iodine
- 59.** The process requiring the absorption of energy is :
- $\text{F} \rightarrow \text{F}^-$
 - $\text{Cl} \rightarrow \text{Cl}^-$
 - $\text{O}^- \rightarrow \text{O}^{2-}$
 - $\text{H} \rightarrow \text{H}^-$
- 60.** The places that were left empty by Mendeleef were, for -
- Aluminium & Silicon
 - Gallium and germanium
 - Arsenic and antimony
 - Molybdenum and tungsten
- 61.** Photoelectric emission is observed from a surface for frequencies ν_1 and ν_2 of incident radiations ($\nu_1 > \nu_2$). If the maximum K.E. of photoelectrons in two cases are in the ratio of 2 : 1, then threshold frequency ν_0 is given by -
- $\frac{\nu_2 - \nu_1}{2}$
 - $\frac{2\nu_1 - \nu_2}{(2-1)}$
 - $\frac{2\nu_2 - \nu_1}{(2-1)}$
 - $\nu_2 - \nu_1$
- 62.** Which excited state of Be^{3+} has the same orbit radius as that of the ground state of hydrogen atom ?
- 3
 - 2
 - 4
 - 1
- 63.** Number of electrons present in $_{17}\text{Cl}$ with $l = 0$
- 6
 - 8
 - 10
 - 12
- 64.** For which of the following species, Bohr's theory is not applicable ?
- Be^{3+}
 - Li^{2+}
 - He^{2+}
 - H
- 65.** The number of mole of KCl in 1000 mL of 3 molar solution is:
- 1.5
 - 3.0
 - 1.0
 - 4.0
- 66.** Which of the following contains largest number of moles
- 1.0 g of O atoms
 - 1.0 g of O_2
 - 1.0 g of NaCl
 - 1.0 g of HCl

- 67.** Second electron gain enthalpy ?
 (1) is always negative
 (2) is always positive
 (3) can be positive or negative
 (4) is always zero
- 68.** Which of the following is not isoelectronic series ?
 (1) Cl^- , P^{3-} , Ar
 (2) N^{3-} , Ne, Mg^{2+}
 (3) B^{3+} , He, Li^+
 (4) N^{3-} , S^{2-} , Cl^-
- 69.** Which of the following electronic configurations in the outermost shell is characteristic of alkali metals?
 (1) $(n-1)s^2ns^2np^1$
 (2) $(n-1)s^2p^6d^{10}ns^1$
 (3) $(n-1)s^2p^6ns^1$
 (4) $ns^2np^6(n-1)d^{10}$
- 70.** Select order in which hydra acid of halogen family are arranged in increasing order of acidic strength.
 (1) HF, HCl, HBr, HI
 (2) HI, HBr, HCl, HF
 (3) HF, HCl, HI, HBr
 (4) HF, HBr, HCl, HI
- 71.** The ionic radii (in Å) of N^{3-} , O^{2-} and F^- are respectively :-
 (1) 1.71, 1.40 and 1.36
 (2) 1.71, 1.36 and 1.40
 (3) 1.36, 1.40 and 1.71
 (4) 1.36, 1.71 and 1.40
- 72.** 4th shell of atom contains 4s, 4p, 4d and 4f subshell, which subshells are not filled in any 4th period element?
 (1) 4p, 4d, 4f (2) 4s, 4p
 (3) 4p, 4f (4) 4d, 4f
- 73.** The second line in Balmer series correspond to an electronic transition between which Bohr's orbits in hydrogen :-
 (1) $4 \rightarrow 2$ (2) $5 \rightarrow 2$
 (3) $5 \rightarrow 3$ (4) $3 \rightarrow 1$
- 74.** The simplest formula of a compound containing 50% of element X (atomic mass 10) and 50% of element Y (atomic mass 20) is :-
 (1) XY (2) X_2Y
 (3) XY_3 (4) X_2Y_3
- 75.** The volume occupied by 4.6 gm of NO_2 at STP is :-
 (1) 22.4 L (2) 2.24 L
 (3) 0.224 L (4) 0.1 L
- 76.** 10.8 kg of silver was recovered from a. Smuggler. How many atoms of silver were recovered.
 (Ag = 108)
 (1) 100 (2) 6.02×10^{23}
 (3) 6.02×10^{24} (4) 6.02×10^{25}
- 77.** Which is not deflected by electric field?
 (1) Proton
 (2) Neutron
 (3) Electron
 (4) All of the above
- 78.** The shape of an orbital is given by the quantum number
 (1) n (2) l
 (3) m (4) s
- 79.** Cation is _____ than its parent atom.
 (1) larger (2) similar
 (3) smaller (4) unpredictable
- 80.** What is the outermost electronic configuration of Pt :-
 (1) $5d^{10}6s^0$ (2) $5d^96s^1$
 (3) $5d^86s^2$ (4) $4d^95s^1$

- 81.** Which one of the following is not a constituent of cell membrane?
 (1) Phospholipids
 (2) Cholesterol
 (3) Glycolipids
 (4) Proline
- 82.** The two subunits of ribosome remain united at a critical ion level of :-
 (1) Magnesium (2) Calcium
 (3) Copper (4) Manganese
- 83.** The main arena of various types of activities of a cell is :-
 (1) Plasma membrane
 (2) Mitochondrion
 (3) Cytoplasm
 (4) Nucleus
- 84.** Match the following and select the correct answer :-

(A) Centriole	(i) Infoldings in mitochondria
(B) Chlorophyll	(ii) Thylakoids
(C) Cristae	(iii) Nucleic acids
(D) Ribozymes	(iv) Basal body of cilia or flagella

 (1) A-(iv) B-(ii) C-(i) D-(iii)
 (2) A-(i) B-(ii) C-(iv) D-(iii)
 (3) A-(i) B-(iii) C-(ii) D-(iv)
 (4) A-(iv) B-(iii) C-(i) D-(ii)
- 85.** The osmotic expansion of a cell kept in water is chiefly regulated by :-
 (1) Mitochondria
 (2) Vacuoles
 (3) Plastids
 (4) Ribosomes
- 86.** Cellular organelles with membranes are :-
 (1) Lysosomes, Golgi apparatus and mitochondria
 (2) Nuclei, ribosomes and mitochondria
 (3) Chromosomes, ribosomes and endoplasmic reticulum
 (4) Endoplasmic reticulum, ribosomes and nuclei
- 87.** Nuclear envelope is a derivative of :-
 (1) Rough endoplasmic reticulum
 (2) Smooth endoplasmic reticulum
 (3) Membrane of Golgi complex
 (4) Microtubules
- 88.** Which of the following components provides sticky character to the bacterial cell?
 (1) Nuclear membrane
 (2) Plasma membrane
 (3) Glycocalyx
 (4) Cell wall
- 89.** Select the correct matching in the following pairs :-
 (1) Rough ER – oxidation of fatty acids
 (2) Smooth ER – oxidation of phospholipids
 (3) Smooth ER – synthesis of lipids
 (4) Rough ER – synthesis of glycogen
- 90.** Keeping in view the "fluid mosaic model" for the structure of cell membrane, which one of the following statements is correct with respect to the movement of lipids and proteins from one lipid monolayer to the other (described as flip-flop movement)?
 (1) While proteins can flip-flop, lipid cannot
 (2) Neither lipids, nor proteins can flip-flop
 (3) Both lipids and proteins can flip-flop
 (4) While lipids can rarely flip-flop, proteins cannot
- 91.** In plant cytokinesis, cell plate grows :-
 (1) Centripetally
 (2) Centrifugally
 (3) Atrandom
 (4) Terminally
- 92.** Term meiosis was coined by :-
 (1) Flemming
 (2) Strasburger
 (3) Farmer and Moore
 (4) Remack

- 93.** Stages in proper sequence of prophase I are :-
 (1) Leptotene, Zygotene, Pachytene, Diplotene, Diakinesis
 (2) Leptotene, Pachytene, Diplotene, Zygotene, Diakinesis
 (3) Leptotene, Diplotene, Diakinesis, Pachytene, Zygotene
 (4) Leptotene, Zygotene, Pachytene, Diakinesis, Diplotene
- 94.** Crossing over occurs during :-
 (1) Diplotene
 (2) Pachytene
 (3) Zygotene
 (4) Leptotene
- 95.** In Meiosis the homologous chromosomes are separated during :-
 (1) Metaphase I
 (2) Prophase II
 (3) Anaphase I
 (4) Anaphase II
- 96.** Match column I and column II and mark the correct set of option :-

Column I		Column II	
(A)	Leptotene	(i)	Appearance of chiasmata
(B)	Zygotene	(ii)	Crossing over
(C)	Pachytene	(iii)	Condensation of chromatin
(D)	Diplotene	(iv)	Formation of synaptonemal complex

- (1) A-(iii) B-(iv) C-(ii) D-(i)
 (2) A-(iii) B-(i) C-(ii) D-(iv)
 (3) A-(ii) B-(iii) C-(iv) D-(i)
 (4) A-(iii) B-(i) C-(iv) D-(ii)

- 97.** The anaphase II is marked by :-
 (1) Simultaneous separation of homologous chromosome
 (2) Splitting of centromere and separation of sister chromatid
 (3) Formation of double equatorial plate
 (4) Appearance of recombination nodule
- 98.** Bouquet stage can be observed in :-
 (1) Leptotene (2) Zygotene
 (3) Pachytene (4) Diplotene

- 99.** In meiosis :-
 (1) First division is reductional and second division is equational
 (2) First division is equational and second division is reductional
 (3) Both divisions are reductional
 (4) Both divisions are equational
- 100.** Chromosome number is reduced to half in :-
 (1) Equational division
 (2) Meiosis II
 (3) Homotypic division
 (4) Meiosis I
- 101.** Cells of squamous epithelium are :-
 (1) Tall with elongated nuclei
 (2) Flat plate-like
 (3) Cubical in shape
 (4) Columnar
- 102.** The epithelium found in the lining of stomach and intestine is :-
 (1) Squamous (2) Columnar
 (3) Transitional (4) Pseudostratified
- 103.** Squamous epithelium is also called :-
 (1) Columnar epithelium
 (2) Pavement epithelium
 (3) Sensory epithelium
 (4) Germinal epithelium
- 104.** The tissue has single thin layer of flattened cells, have irregular boundaries, flattened nuclei and found in the wall of blood vessels. All these features can be related to :-
 (1) Simple columnar epithelium
 (2) Transitional epithelium
 (3) Simple cuboidal epithelium
 (4) Simple squamous epithelium
- 105.** Which of the following is uncommon for epithelial tissue?
 (1) Cells are held together by cell junctions
 (2) Highly vascular
 (3) Cells are compately packed
 (4) Presence of basement membrane

- 106.** Cells of germinal epithelium are
(1) Columnar (2) Ciliated
(3) Squamous (4) Cuboidal
- 107.** An example of merocrine gland is :-
(1) Sweat gland (2) Sebaceous gland
(3) Mammary gland (4) All of these
- 108.** Haversian canals are found in :-
(1) Adipose tissue (2) Bone
(3) Muscle (4) Cartilage
- 109.** Bone forming cells are :-
(1) Osteoclasts (2) Chondroblasts
(3) Chondroclasts (4) Osteoblasts
- 110.** In a bone, concentric layer of matrix are called :-
(1) Lamellae
(2) Volkmann's canal
(3) Lacunae
(4) Haversian canals
- 111.** Cellulose and inulin differ in :-
(1) Type of glycosidic linkage
(2) Functions being served
(3) Type of monomer unit
(4) All of these
- 112.** In human, homopolymer of glucose is stored as :-
(1) Starch and glycogen in liver
(2) Glycogen in liver and muscles
(3) Lactic acid in muscles
(4) Glycogen in liver only
- 113.** Match the following :-

(A)	GLUT-4	(i)	Fungal cellulose
(B)	Collagen	(ii)	Intercellular ground substance
(C)	RUBISCO	(iii)	Most abundant protein in biosphere
(D)	Chitin	(iv)	Enables glucose transport into cell

- (1) A-(iv), B-(ii), C-(iii), D-(i)
(2) A-(iv), B-(iii), C-(ii), D-(i)
(3) A-(ii), B-(iv), C-(i), D-(iii)
(4) A-(i), B-(iv), C-(ii), D-(iii)

- 114.** The semi-essential amino acids are :-
(1) glycine and arginine
(2) arginine and histidine
(3) lysine and tryptophan
(4) serine and alanine
- 115.** Which of the following bonds does not break down during denaturation of protein?
(1) Hydrogen bond
(2) Ionic bond
(3) Hydrophobic bond
(4) Peptide bond
- 116.** All of the following are saturated fatty acids, except :-
(1) Palmitic acid
(2) Stearic acid
(3) Arachidic acid
(4) Linolenic acid
- 117.** Which of the following nitrogen base is not found in DNA?
(1) Adenine (2) Thymine
(3) Uracil (4) Guanine
- 118.** The pitch of B-DNA is :-
(1) 3.4 Å (2) 34 Å
(3) 20 Å (4) 0.34 nm
- 119.** In eukaryotes, DNA is found in :-
(1) Nucleus
(2) Mitochondria
(3) Plastids
(4) All of these
- 120.** Find the correct match :-

Column I		Column II	
(A)	Polymer	(i)	Abrin
(B)	Alkaloid	(ii)	Anthocyanin
(C)	Toxin	(iii)	Cellulose
(D)	Pigment	(iv)	Morphine

- (1) A-(i), B-(iii), C-(ii), D-(iv)
(2) A-(i), B-(ii), C-(iii), D-(iv)
(3) A-(iii), B-(iv), C-(i), D-(ii)
(4) A-(iii), B-(i), C-(iv), D-(ii)

Direction for Assertion & Reason questions. These questions consist of two statements each, printed as assertion and reason. while answering these questions you are required to choose any one of the following four responses.

- (1) If both Assertion & Reason are true & the reason is a correct explanation of the assertion.
- (2) If both Assertion & Reason are true but reason is not a correct explanation of the assertion.
- (3) If Assertion is true but the Reason is false
- (4) If both Assertion & Reason are false

121. Assertion :- Vector $(\hat{i} + \hat{j} + \hat{k})$ is perpendicular to $(\hat{i} - 2\hat{j} + \hat{k})$.

Reason :- Two non-zero vectors are perpendicular if their dot product is zero.

122. Assertion :- Distance is a scalar quantity.

Reason :- Distance is the length of path traversed

123. Assertion :- The dot product of one vector with another vector may be a scalar or a vector.

Reason :- If the product of two vectors is a vector quantity, then product is called a dot product

124. Assertion :- The angle between the two vectors

$(\hat{i} + \hat{j})$ and $(\hat{j} + \hat{k})$ is $\frac{\pi}{3}$ radian.

Reason :- Angle between two vectors $\vec{A}$ and

$\vec{B}$ is given by $\theta = \cos^{-1} \left(\frac{\vec{A} \cdot \vec{B}}{AB} \right)$

125. Assertion :- A vector can have zero magnitude if one of its rectangular components is not zero.

Reason :- Scalar product of 2 vectors cannot be a negative quantity.

126. Assertion :- A vector quantity is a quantity that has both magnitude and a direction and obeys the triangle law of addition and equivalently the parallelogram law of addition.

Reason :- The magnitude of the resultant vector of two given vectors can never be less than the magnitude of any of the given vector.

127. Assertion :- If the initial and final positions coincide, the displacement is a null vector.

Reason :- A physical quantity cannot be called a vector, if its magnitude is zero.

128. Assertion :- $(\vec{A} \times \vec{B}) \cdot (\vec{B} \times \vec{A})$ is $-A^2 B^2 \sin^2 \theta$.

Here θ is the angle between $\vec{A}$ and $\vec{B}$.

Reason :- $(\vec{A} \times \vec{B})$ and $(\vec{B} \times \vec{A})$ are two anti-parallel vectors provided A and B are neither parallel nor anti-parallel

129. Assertion :- $\vec{A} \times \vec{B}$ is perpendicular to both $\vec{A} + \vec{B}$ as well as $\vec{A} - \vec{B}$.

Reason :- $\vec{A} + \vec{B}$ as well as $\vec{A} - \vec{B}$ lie in the plane containing $\vec{A}$ and $\vec{B}$ while $\vec{A} \times \vec{B}$ lies perpendicular to the plane containing $\vec{A}$ and $\vec{B}$.

130. Assertion :- If angle between $\vec{a}$ and $\vec{b}$ is 30° .

Then angle between $2\vec{a}$ and $-\frac{\vec{b}}{2}$ will be 150° .

Reason :- Sign of dot product of two vectors tells you whether angle between two vectors is acute or obtuse

131. Assertion :- Angle between $\hat{i} + \hat{j}$ and $\hat{i}$ is 45° .

Reason :- $\hat{i} + \hat{j}$ is equally inclined to both $\hat{i}$ and $\hat{j}$ and the angle between $\hat{i}$ and $\hat{j}$ is 90° .

132. Assertion :- Small displacement is a vector quantity.

Reason :- Mass and distance are also vector quantities

- 133. Assertion :-** The sum of squares of cosines of angles made by a vector with X, Y and Z-axis is equal to unity.
Reason :- A vector making 45° with x-axis must have equal components along X and Y-axis
- 134. Assertion :-** Adding a scalar to a vector of the same dimensions is a meaningful algebraic operation.
Reason :- Displacement can be added to distance.
- 135. Assertion :** The equation of motion can be applied only if the acceleration is along the direction of velocity and is constant.
Reason : If the acceleration of a body is constant then its motion is known as uniform motion.
- 136. Assertion :** Velocity-time graph of an object in uniform motion along a straight path is a straight line parallel to the time axis.
Reason : In uniform motion of an object, velocity increases as the square of time elapsed.
- 137. Assertion :** Two balls of different masses are thrown vertically upward with same speed. They will pass through their point of projection in the downward direction with the same speed.
Reason : The maximum height and downward velocity attained at the point of projection are independent of the mass of the ball.
- 138. Assertion :** The average speed of an object may be equal to arithmetic mean of individual speeds.
Reason : Average speed is equal to total distance travelled per total time taken.
- 139. Assertion :** The average and instantaneous velocities have same value in a uniform motion.
Reason : In uniform motion, the velocity of an object increases uniformly.
- 140. Assertion :** The speedometer of an automobile measures the average speed of the automobile.
Reason : Average velocity is equal to total displacement per total time taken.
- 141. Assertion :** Volume occupied by 1 mol NaOH(s) is equal to 22400 cc at NTP.
Reason : 1 mol of any substance occupied 22.4L volume at NTP.
- 142. Assertion :** M^{+3} (at. no. 26) has magnetic moment (spin only) $\sqrt{35}$.
Reason : M^{+3} contains 5 unpaired electron.
- 143. Assertion :** Bohr model is applicable for multielectronic species.
Reason : We can find out radius of He, Li^+ , Be^{+2} , with the help of Bohr model.
- 144. Assertion :** Number of radial and angular node for 4d orbital are 1, 2 respectively.
Reason : No. of radial and angular node depends only on principal quantum number.
- 145. Assertion :** Fluorine is most electronegative element.
Reason : Fluorine has tendency to lose electrons.
- 146. Assertion :** Ne and Na are chemically inert.
Reason : Ne and Na have inert gas electronic configuration.
- 147. Assertion :** The 1st IP of Be is greater than that of B.
Reason : Full filled electronic configuration is more stable as compared to half-filled and partially filled electronic configuration.
- 148. Assertion :** Atomic size along a period generally decreases.
Reason : Z_{eff} in period increases.
- 149. Assertion :** Stable electronic configuration affects electronegativity.
Reason : EN is the tendency of an atom to gain electrons.
- 150. Assertion :** Properties of Beryllium are similar to that of aluminium.
Reason : 2nd period elements show diagonal relationship with 3rd period elements.
- 151. Assertion :** Second IP of oxygen is greater than that of fluorine.
Reason : Oxygen acquires stable half-filled electronic configuration after losing one electron.

152. Assertion : The mass of the product formed in a reaction depends upon the excess reactant.
Reason : Excess reactant reacts completely in the reaction.

153. Assertion : Carbon and oxygen combined together only in one fixed ratio.

Reason : Law of conservation of mass holds good for nuclear reaction.

154. Assertion : Nodal plane of P_y atomic orbital is the XZ plane.

Reason : In P_y atomic orbital, electron density is zero in the XZ plane.

155. Assertion : The angular momentum of d-orbitals is $\sqrt{6} \frac{h}{2\pi}$.

Reason : Angular momentum of electron orbits

$$\text{is } mvr = \frac{nh}{2\pi}.$$

156. Assertion : Emitted radiation will fall in visible range when an electron jumps from $n = 4$ to $n = 2$ in H-atom.

Reason : Balmer series radiations belong to visible range for hydrogen atom.

157. Assertion : The ground state configuration of Cr is $3d^5 4s^1$.

Reason : A set of half-filled orbitals containing one electron each with their spins parallel provides extra stability.

158. Assertion : The ground state electron configuration of nitrogen is :



Reason : Electrons are filled in orbitals as per aufbau principle, Hund's rule of maximum spin multiplicity and Pauli's principle.

159. Assertion : An orbital cannot have more than two electrons and they must have opposite spins.

Reason : No two electrons in an atom can have the same set of all the four quantum numbers as per Pauli's exclusion principle.

160. Assertion : Orbital having XZ plane as node may be $3d_{xy}$.

Reason : $3d_{xy}$ has zero radial node.

161. Assertion : Mitochondria are called power house of cell.

Reason : It produces energy in respiration which is stored in the form of ATP.

162. Assertion : In plasma membrane extrinsic proteins are abundant towards the cytosolic face.

Reason : Proteins are arranged asymmetrically and show flip-flop movement.

163. Assertion : Membrane transport occurs through carrier proteins.

Reason : The transport carried by carrier proteins is always active.

164. Assertion : Mitochondria and chloroplast are believed to be bacterial endosymbionts of cells.

Reason : They have their own nucleic acid and ribosome.

165. Assertion : Eukaryotic cells have more DNA than prokaryotes.

Reason : In prokaryotic cell genetic material is DS circular DNA without histone protein.

166. Assertion : Morphology of the chromosomes is most easily studied during metaphase.

Reason : Condensation of chromosomes is completed in metaphase.

167. Assertion : Prophase of meiosis I is typically longer and more complex compared to the prophase of mitosis.

Reason : In contrast to meiosis I, meiosis II prophase resembles a normal mitosis.

168. Assertion : The complex formed by a pair of synapsed homologous chromosomes is called a "bivalent" or a tetrad.

Reason : The first two stages of meiosis prophase I are relatively long-lived compared to the next stage, "pachytene".

169. Assertion : Diakinesis is marked by terminalization of chiasmata.

Reason : During this phase, the chromosomes are fully condensed and the meiotic spindle is assembled to prepare the homologous chromosome for separation.

170. Assertion : The homologous chromosomes separate during anaphase I.

Reason : During anaphase I, sister chromatids remain associated at their centromeres.

- 171. Assertion :** The process of bone formation is called ossification
Reason : Osteoclasts are bone forming cells.
- 172. Assertion :** Hyaline is called articular cartilage.
Reason : Hyaline cartilage covers the articular surfaces of long bones in joints.
- 173. Assertion :** Cartilage does not have rich blood supply.
Reason : Cartilage injuries heal easily.
- 174. Assertion :** Simple cuboidal epithelium is also called as Germinal epithelium.
Reason : cuboidal cells of gonads forms gametes.
- 175. Assertion :** Glands originate from all germinal layers.
Reason : All glands are composed of epithelium tissue.

- 176. Assertion :** Cellulose is a homopolymer, while hemicellulose is a heteropolymer.
Reason : Cellulose is made up of identical monomers while hemicellulose is made up of variable monomer units.
- 177. Assertion :** Lactose is a reducing sugar.
Reason : In lactose 1-4 glycosidic linkage is present, so free $-CHO$ group present, which is responsible for reducing nature.
- 178. Assertion :** All disaccharides are reducing sugars.
Reason : disaccharides contain free aldehyde or free ketonic group.
- 179. Assertion :** Phospholipid form biomolecular layer in aqueous medium.
Reason : Phospholipid molecules are amphipathic.
- 180. Assertion :** Adenine and uracil are complementary bases of DNA.
Reason : Uracil is purine base.



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